what an egg-unscrambling machine would do to its surroundings, but I'm pretty sure I wouldn't want to be in a closed room with it when it's running.

This, incidentally, is why leaving your refrigerator door open will ultimately make the whole kitchen hotter, and why air conditioners can contribute to global warming. Every attempt to bend some part of the world to our will creates disorder somewhere else, often in the form of heat.

As much as this has interesting applications for eggs and fridges and air conditioners, it all gets *much* weirder when we throw black holes into the mix.

Back in the 1970s, physicists were talking a lot about entropy and how the entropy of the *whole universe* must be increasing over time, and what the implications of that might be. At the same time, a young, not-quite-famous-yet Stephen Hawking and even younger postdoctoral researcher Jacob Bekenstein were thinking about black holes and wondering if these bizarre, inescapable, spacetime garbage disposals might wreak havoc on the Second Law of Thermodynamics. What if, for example, you used your egg-unscrambler to unscramble the egg, and then pocketed the egg while throwing the whole messy hot unscrambler lab into the nearest black hole? Would you have decreased the overall entropy of the universe by putting the egg back together and getting rid of all the entropy you created in the process? After all, a black hole is defined as something that not even light can escape from, an object so massive and compact that its gravity turns outgoing light rays right around to send them diving back toward the central singularity. Beyond the event horizon of a black hole, the gravitational point of no return, nothing—not light, not information, not heat—can escape once it's gone in. Could hiding entropy behind black hole event horizons be the perfect crime?

Whatever other part of physics you have to break, don't bet against the Second Law of Thermodynamics. The solution to the entropy problem of black holes turned out to change everything we thought we knew about black holes and absolutely nothing about entropy. You can't hide entropy in black holes, because they have entropy of their own. Which means they have a temperature (they create heat). Which means they are not black at all.

What Bekenstein and Hawking eventually concluded about black holes is that a black hole has to have an entropy associated with it, to exist in accordance with the Second Law. Since that entropy should increase every